

Spine — knowledge base for Otto

Everything the help agent needs to answer questions about the Spine book scanner. Written in plain language so it can be read aloud.

About this document

This file is the primary knowledge base for "Otto", the archivist owl who answers user questions inside the Spine app by text and by voice. It is generated from the same FAQ file the app ships with, so answers here match what users read on the Help page. Upload it to the agent's knowledge base in ElevenLabs. Companion PDFs for deeper technical topics are listed in the final section with links to their sources.

Who Otto is and how he speaks

- Otto is a pixel-art owl, the archivist of the user's bookshelf. Warm, brief, a little dry, never sarcastic.
- Answers in two to four short sentences. Offers one concrete next step. Does not read long lists aloud.
- Uses the names the app uses: Shelf, Scan, Reading tab, Cover text, To read, Reading, Finished, Prepare for offline, Get the app.
- If a question is outside Spine (general book recommendations, unrelated apps), says so kindly and steers back to the shelf.
- Never asks for passwords, payment details or personal information. Spine has no accounts.
- If unsure, says "I am not certain" and suggests the Help page in Settings rather than guessing.

Product facts

Name	Spine
What it is	A progressive web app that scans book barcodes and covers with the phone camera and files each book on a personal shelf with its catalogue record.
Web address	https://sultan-alkaabi-hub.github.io/app/
Platforms	Any modern browser. Installable on Android (Chrome "Install app", or an APK) and iPhone (Safari "Add to Home Screen").
Version	1.2.0
Data source	Open Library, a free public catalogue run by the Internet Archive. No API key.
Accounts	None. Nothing is uploaded. The shelf is stored in the browser on the device.
Barcode engines	The phone's built-in barcode detector where available ("On-device engine"); otherwise a JavaScript decoder downloaded once ("Web engine").
Cover reading	Optical character recognition on the front cover, then a shortlist of matching editions for the user to confirm.
Valid ISBNs	Start with 978 or 979, 10 or 13 digits, and pass the check digit. 977 is a magazine. Two or five digit codes are price add-ons.
Offline	App opens offline. Barcode scanning works offline on phones with a built-in detector. Lookups need internet; offline scans are saved as pending books and resolved on reconnect.

Reading statuses	To read, Reading, Finished.
Views	Cover grid, or spines standing on wooden shelves (widths from page count, colours sampled from covers).
Statistics	Books on shelf, finished, pages read, authors, status breakdown, top authors, subjects, decades of publication.
Backup	Settings → Export your shelf writes a JSON file. Import merges by ISBN.
Timeouts	Open Library can take 5–12 seconds cold. The app waits up to 20 seconds, then keeps the scan as pending.

Step by step: how the app is used

First launch

- The intro screen shows a lamp-lit bookshelf with three buttons: Start scanning, Show me how it works, and Get the app on your phone.
- Show me how it works starts Otto's five-step walkthrough: two ways to scan, the barcode, the cover, where the data comes from, and what happens offline.
- Start scanning opens the walkthrough the first time; afterwards it opens the shelf, or the scanner if the shelf is empty.

Scanning a barcode

- Tap the round scan button in the middle of the bottom bar. The camera opens with a frame in the middle.
- The first time, the browser asks for camera permission. Allow it.
- Flip the book over and hold the barcode inside the frame. The corners turn green and the phone vibrates when it locks on.
- Spine fetches the record and opens the book sheet: cover, title, author, publisher, pages, year, status buttons, blurb, subjects, and a link to the catalogue record.
- If the book is already on the shelf, Spine says so and opens it instead of adding a duplicate.

Scanning a cover

- On the scanner, tap Cover text in the switch at the bottom. The frame becomes portrait.
- Fill the frame with the front cover and tap the round white button. The first time, the text reader downloads (a few megabytes).
- Spine reads the printed text and shows a shortlist of matching editions with covers. Tap the right one. Or close it and use the barcode instead.

Typing an ISBN

- Tap the keypad icon at the top right of the scanner, or "Type an ISBN instead" on any camera error.
- Enter the 10 or 13 digit number. Spine explains any problem: wrong check digit, magazine code, price add-on, product barcode.

The shelf

- Books appear newest first. Filter chips: All, Reading, To read, Finished.
- The magnifier searches title, author, subject and ISBN. The button beside it toggles the spine view.
- Tap a book to open it. Change status, read the blurb, open the catalogue record, or remove it.
- Pending books (scanned offline) show "ISBN" on a striped jacket and are looked up automatically when online.

Settings

- Install Spine (when the browser offers it), Get the app on another phone (QR code and links), Prepare for offline, Vibrate on a match, Reduce motion, Help & FAQ, Ask Otto, Replay the walkthrough, Export, Import, Delete everything.
- A new-version pill with a Reload button appears when a newer build is published.

Asking Otto

- The Otto button at the bottom right of the shelf, Reading and Settings pages opens the chat.
- Type a question, or tap the microphone and speak. Otto replies in text and reads it aloud; the speaker icon mutes him.

Troubleshooting quick reference

Symptom	Cause	What to say
"This page is not secure"	Opened over http	Open the https address. Camera access needs HTTPS.
"Camera permission is off"	Permission denied in browser	Android: lock icon → Permissions → Camera → Allow. iPhone: Settings → Safari → Camera. Then Try again.
"This app build cannot use the camera"	Installed package built without camera permission	Phone Settings → Apps → Spine → Permissions → Camera. If no Camera entry, the app must be rebuilt with the permission. Use the browser meanwhile.
"No camera found"	Device has no camera the browser can reach	Type the ISBN instead.
"The camera is busy"	Another app or tab holds the camera	Close other camera apps and tabs, wait a second, Try again.
Barcode rejected as magazine / price / product	Wrong barcode	Aim at the long barcode starting 978 or 979.
Lookup slow or timed out	Open Library cold request	Wait; the scan is kept as pending and retried. Or "Try the lookup again" inside the book.
No cover / blurb / pages	Gaps in Open Library	Normal for some editions. The catalogue link lets users add details.
Cover scan wrong book	OCR misread	Pick from the shortlist, or scan the barcode. More light, flat angle.
Offline pill at the top	No connection	Scanning still works; records arrive when back online.

Frequently asked questions

Identical to the Help page inside the app. Each answer is written to be spoken.

Getting started

Q: What is Spine?

Spine is a book scanner and personal shelf. Point your phone camera at a book and Spine reads the barcode or the cover, looks the book up in the Open Library catalogue and files it on your shelf with its pages, publisher, subjects, blurb and cover. It runs in the browser as a progressive web app and can be installed on Android and iPhone.

Also asked as: what is spine, about, what does this app do, purpose

Q: How do I scan a book?

Tap the big scan button in the middle of the bottom bar. Flip the book over and hold the barcode inside the frame. Spine locks on in about a second, fetches the record and opens the book. For books without a barcode, switch to Cover text, fill the frame with the front cover and tap the round button.

Also asked as: how do i scan, scan a book, how to scan, start scanning, add a book

Q: Who is Otto?

Otto is the archivist owl who runs the walkthrough and answers questions. He explains the two ways to scan, what happens offline, and where your data lives. You can replay his walkthrough from Settings, and talk to him from the Ask Otto button on any page.

Also asked as: otto, owl, mascot, who are you

Q: Do I need an account?

No. Spine has no accounts and no server of its own. Your shelf is stored on your device and nothing is uploaded.

Also asked as: account, sign up, login, log in, register

Scanning

Q: Which is better, barcode or cover?

The barcode is exact. It carries the ISBN, which names the precise edition you are holding, and it reads instantly with almost no processing. Cover text uses optical character recognition, which is slower and can misread stylised fonts, so Spine shows you a shortlist to confirm rather than guessing. Use the barcode whenever the book has one.

Also asked as: barcode or cover, which is better, difference, barcode vs cover, cover text

Q: The scanner says the barcode is a magazine or a price code. Why?

Book barcodes start with 978 or 979. A code starting with 977 belongs to a magazine or journal. The short two or five digit barcode printed next to the main one is the price add-on. Aim at the longer barcode. Ordinary product barcodes with 8 or 12 digits are not ISBNs either.

Also asked as: magazine, price, add-on, wrong barcode, not a book, product barcode, 977

Q: Can I type the ISBN instead?

Yes. On the scanner, tap the keypad icon at the top right, or choose Type an ISBN instead on any camera error. Enter the 10 or 13 digit number printed under the barcode. Dashes and spaces are fine, and Spine checks the number against its check digit before looking it up.

Also asked as: type isbn, type the isbn, manual, enter isbn, keyboard, no camera

Q: The cover scan found the wrong book.

Cover reading is a best guess from the printed text, so Spine always shows a shortlist of matches instead of adding one automatically. Pick the right edition from the list, or close it and scan the barcode on the back for an exact match. More light, a flat angle and holding still improve results.

Also asked as: wrong book, cover wrong, misread, ocr wrong, not the right book

Q: How do I turn on the flashlight?

When the camera supports it, a flashlight button appears at the top right of the scanner. Tap it to toggle the torch. If the button is missing, the device or browser does not expose torch control, which is common on iPhone in Safari.

Also asked as: flashlight, torch, dark, light, flash

Q: Which barcode engine is Spine using?

Spine uses the phone's built-in barcode detector when the browser has one, shown as On-device engine at the top of the scanner. It needs no download and works without a signal. Browsers without a built-in detector, such as Safari, get the Web engine, a JavaScript decoder that is downloaded once and then cached.

Also asked as: engine, on-device engine, web engine, badge, barcodedetector

Camera and permissions

Q: The camera does not open.

First check the message on the screen; Spine names the exact problem. Camera access only works on the secure https address, so make sure the address starts with https. If permission was denied earlier, allow it again in the browser or phone settings, then tap Try again. If the camera is busy, close other apps that might be using it. You can always type the ISBN instead.

Also asked as: camera does not open, camera not working, camera unavailable, no camera, black screen

Q: How do I allow camera permission?

On Android Chrome, tap the lock icon in the address bar, open Permissions and set Camera to Allow. On iPhone, open Settings, Safari, Camera and choose Allow, or Ask. In the installed app, open the phone Settings, Apps, Spine, Permissions and allow Camera. Then return to Spine and tap Try again.

Also asked as: permission, allow camera, camera permission, denied, blocked

Q: The installed app never asks for the camera.

That means the app package was built without the camera permission, so the phone cannot prompt for it. The package has to be rebuilt with the camera permission enabled in the app builder. Until then, use Spine in the browser at its https address, or install it from Chrome using Install app.

Also asked as: never asks, apk permission, app build, wrapper, median, no prompt

Q: It says the camera is used by another app.

Another app or browser tab is holding the camera, or the phone is still releasing it from a moment ago. Close other camera apps and tabs, wait a second and tap Try again. Spine retries automatically once before showing this message.

Also asked as: another app, camera busy, in use, could not start

Offline and data

Q: Does Spine work offline?

Mostly, yes. The app itself opens offline once it has loaded once. Barcode scanning works with no signal on phones with a built-in detector. The catalogue lookup needs the internet, so offline scans are saved as pending books showing their ISBN, and Spine fetches the full record the moment you reconnect. Cover reading needs one online run, or the Prepare for offline step in Settings, to download its text reader.

Also asked as: offline, no signal, no internet, airplane, without internet, no connection

Q: What does Prepare for offline do?

It downloads the web barcode engine and the cover text reader once and stores them on the device, so both kinds of scanning work later without a signal. It takes a few megabytes, so do it on Wi-Fi.

Also asked as: prepare for offline, download engines, offline engines

Q: Where is my data stored? Is it private?

Your shelf lives in the browser storage on your own device. Nothing is uploaded, there are no accounts and no analytics. The only network requests are to Open Library, the free public catalogue run by the Internet Archive, to fetch book records and covers.

Also asked as: privacy, data, stored, where is my data, private, upload, tracking

Q: How do I back up or move my shelf?

Open Settings and tap Export your shelf. Spine writes a JSON file you can save or share. On another device, use Import a shelf file. Import merges by ISBN, so nothing is duplicated.

Also asked as: backup, back up, export, import, move, transfer, another phone, json

Q: Where does the book information come from?

From Open Library, a free public catalogue run by the Internet Archive. Coverage is uneven: some editions have no cover, page count or blurb, and Spine says so plainly rather than inventing one. Each book has a link to its catalogue record where you can see or add missing details.

Also asked as: open library, where does the data come from, catalogue, source, internet archive, missing blurb, no cover

Q: The lookup is slow or times out.

Open Library is a volunteer run service and a cold request can take five to twelve seconds. Spine waits up to twenty seconds. If it fails, the scan is kept as a pending book and looked up again later, or you can tap Try the lookup again inside the book.

Also asked as: slow, timeout, timed out, taking long, loading forever

Your shelf

Q: How do reading statuses work?

Open any book and choose To read, Reading or Finished. The filters at the top of the shelf show one status at a time, and the Reading tab counts pages read from your finished books.

Also asked as: status, reading, to read, finished, want to read, mark as read

Q: What is the spine view?

The button beside search switches the shelf between cover tiles and spines standing on wooden shelves. Spine widths come from page counts and colours are sampled from each cover. Tap any spine to open the book.

Also asked as: spine view, spines, shelf view, bookshelf, cover view, grid

Q: How do I search or remove a book?

Tap the magnifier on the shelf to search by title, author, subject or ISBN. To remove a book, open it and tap Remove from shelf at the bottom. Delete everything in Settings clears the whole shelf after a confirmation.

Also asked as: search, find a book, remove, delete a book, delete

Q: What do the statistics show?

The Reading tab shows how many books are on the shelf, how many are finished, pages read, number of authors, a breakdown by status, your most shelved authors, the subjects you read about and the decades your books were published in.

Also asked as: statistics, stats, reading tab, pages read, numbers

Installing

Q: How do I install Spine on Android?

Two ways. Open the site in Chrome and choose Install app from the menu, or tap Get the app on the intro screen for a QR code and a direct APK download. After installing the APK, allow the camera on first scan.

Also asked as: install android, android, apk, download the app, install app, home screen android

Q: How do I install Spine on iPhone?

Open the site in Safari, tap Share, then Add to Home Screen, then Add. Other browsers on iPhone cannot install web apps. The Get the app button shows a QR code with these steps.

Also asked as: install iphone, ios, iphone, add to home screen, safari, apple

Q: There is a Reload pill saying a new version is ready.

Spine caches itself so it opens instantly and offline. When a newer build is published, the pill appears; tap Reload once to switch to it. Your shelf is not affected.

Also asked as: new version, reload, update, latest version

Otto and voice

Q: How do I talk to Otto?

Tap the Otto button at the bottom right of the shelf, Reading or Settings pages. Type a question, or hold the microphone button and speak. Otto answers in text and can read the answer aloud. Tap the speaker icon to turn his voice on or off.

Also asked as: talk, voice, speak, microphone, mic, ask otto, chat

Q: Otto did not understand my question.

Try asking in a few plain words, for example "camera permission" or "does it work offline". You can also open the full FAQ from Settings, where every topic is listed.

Also asked as: did not understand, wrong answer, not helpful

Glossary

ISBN — International Standard Book Number. 13 digits starting 978 or 979 (older books: 10 digits). Printed under the barcode on the back cover. Identifies one edition.

EAN-13 — The barcode format that carries the ISBN on books.

Open Library — Free public book catalogue run by the Internet Archive, at openlibrary.org. Source of all book data in Spine.

PWA — Progressive web app. A website that installs like an app, opens full screen and works offline.

OCR — Optical character recognition. Reading printed text from a photo. Used for cover scanning.

Pending book — A book scanned without internet. Shows its ISBN until the record is fetched.

Service worker — The part of the app that caches it for offline use and shows the Reload pill on updates.

Median — A service that wraps the website in a native Android or iOS app. The wrapper must be built with camera permission.

Companion PDFs: sources to export for the knowledge base

Open each link, print or save the page as PDF, and upload it alongside this document. The first group covers the app itself; the rest give the agent accurate background on the technologies users ask about.

The app itself

- Spine live site (open, then print to PDF the intro, the Help page and Settings)
<https://sultan-alkaabi-hub.github.io/app/>
- Spine README with deployment, Median and offline details
<https://github.com/Sultan-AlKaabi-hub/app#readme>
- Spine source repository
<https://github.com/Sultan-AlKaabi-hub/app>

Book data and ISBNs

- Open Library developer documentation
<https://openlibrary.org/developers/api>
- Open Library Books API (ISBN lookups)
<https://openlibrary.org/dev/docs/api/books>
- Open Library Search API
<https://openlibrary.org/dev/docs/api/search>
- Open Library Covers API
<https://openlibrary.org/dev/docs/api/covers>
- ISBN user manual, International ISBN Agency
<https://www.isbn-international.org/content/isbn-users-manual>
- EAN-13 and Bookland explained (GS1)
<https://www.gs1.org/standards/barcodes/ean-upc>

Camera and scanning technology

- MDN: BarcodeDetector API
<https://developer.mozilla.org/en-US/docs/Web/API/BarcodeDetector>
- MDN: getUserMedia and camera permissions
<https://developer.mozilla.org/en-US/docs/Web/API/MediaDevices/getUserMedia>
- html5-qrcode (web barcode engine) documentation
<https://scanapp.org/html5-qrcode-docs/>
- Tesseract.js (cover text reader) documentation
<https://github.com/naptha/tesseract.js#readme>
- MDN: Web Speech API (Otto's browser voice fallback)
https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API

Installing and offline

- web.dev: Installing a PWA on Android and desktop
<https://web.dev/learn/pwa/installation>
- Apple: Add a website to the Home Screen on iPhone
<https://support.apple.com/guide/iphone/bookmark-favorite-webpages-iph42ab2f3a7/ios>

- Chrome help: Allow or block camera and microphone for a site
<https://support.google.com/chrome/answer/2693767>
- Apple: Control access to the camera in Safari
<https://support.apple.com/guide/iphone/control-access-to-hardware-features-iph1b3dc7a2b/ios>
- MDN: Service workers and offline caching
https://developer.mozilla.org/en-US/docs/Web/API/Service_Worker_API

App wrappers

- Median.co documentation
<https://median.co/docs>
- Median: Camera and permissions
<https://median.co/docs/permissions>
- GitHub Pages documentation
<https://docs.github.com/en/pages>

ElevenLabs agent setup

- ElevenLabs Conversational AI: knowledge base
<https://elevenlabs.io/docs/conversational-ai/customization/knowledge-base>
- ElevenLabs: embedding the widget
<https://elevenlabs.io/docs/conversational-ai/customization/widget>
- ElevenLabs JavaScript client SDK
<https://elevenlabs.io/docs/conversational-ai/libraries/java-script>

Suggested agent system prompt

You are Otto, the archivist owl in the Spine book-scanner app. Answer only from the knowledge base. Keep replies to two to four spoken sentences and end with one concrete next step when there is a problem. Use the app's own names for buttons and pages. If the question is not about Spine, say so kindly in one sentence and offer to help with scanning, the shelf, offline use or installation. Never ask for personal information. If unsure, say you are not certain and point to Settings, Help & FAQ.